EECS 245, Fall 2026

LEC 20 Eigenvalues and Eigenvectors

→ Read: Ch. 9.1 - 9.2

Agenda

- Recap: convexity, strict convexity, tangent hyperplanes

in scope ↑
not in scope ↓ Ch. 9

- Introduction to **eigen**-stuff
  - eigenvalues
  - eigenvectors
  - eigenspaces
  - eigenbases

Announcements

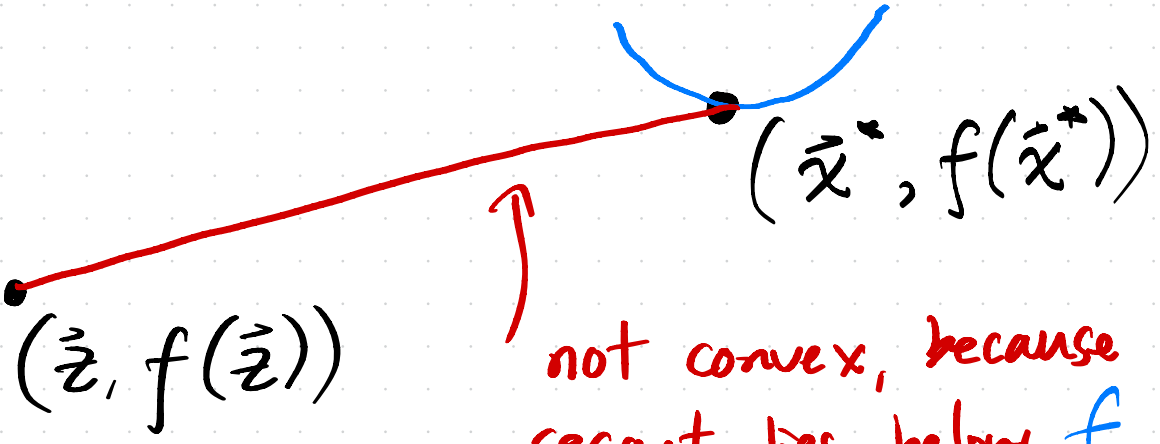
- Midterm 2 on Tuesday from 7-9PM in 1013 DOW! (see logistics on Ed)
- Lots of in-person + Zoom OH, including over weekend (North and Central!)
- Lecture on day of exam: OH
- Summaries posted for Ch. 5-8
- HW 8 due Sunday: no slip days!
- No lab on Wednesday

Convexity

1. Why convexity implies local minimums are global
2. Strict convexity and mean squared error
3. Tangent hyperplanes

Why is a local minimum guaranteed to be a global min, for a convex function f?

→ Suppose $\vec{x}^*$ local min, but $f(\vec{z}) < f(\vec{x}^*)$



not convex, because secant lies below f!

→ Impossible!
Why?

$$R_{sq}(\vec{w}) = \frac{1}{n} \|\vec{y} - X\vec{w}\|^2$$

multiple linear regression $R_{sq} : \mathbb{R}^{d+1} \to \mathbb{R}$

$R_{sq}(\vec{w})$ is convex!

Case (1): X's cols are linearly independent

Unique minimizer: $\vec{w}^* = (X^T X)^{-1} X^T \vec{y}$

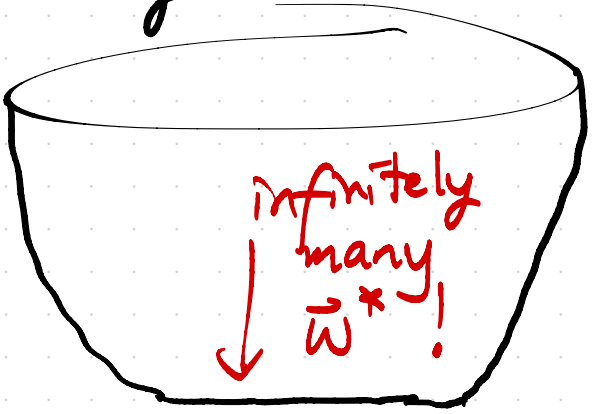
Case (2): X's cols are <u>not</u> linearly independent

Infinitely many minimizers, satisfy $X^T X \vec{w}^* = X^T \vec{y}$

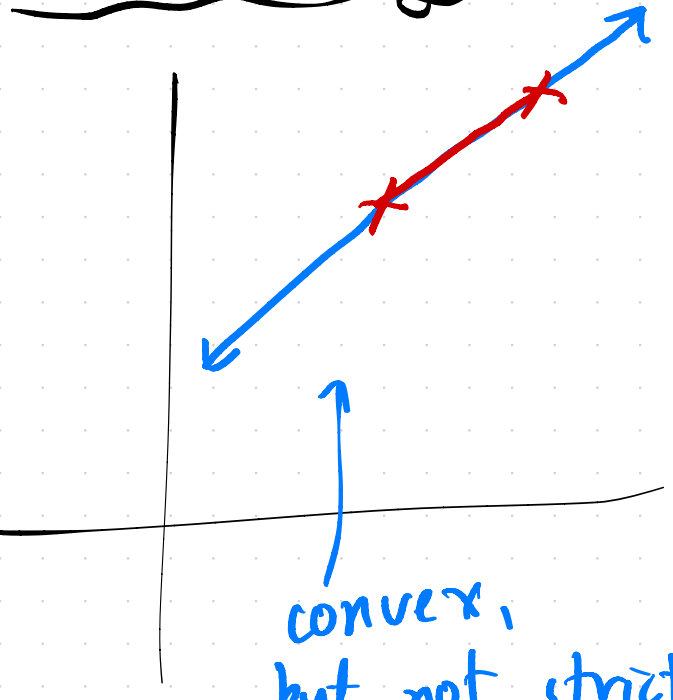
When X's cols are linearly dependent, then

$$R_{sq}(\vec{w}) = \frac{1}{n} \|\vec{y} - X\vec{w}\|^2$$

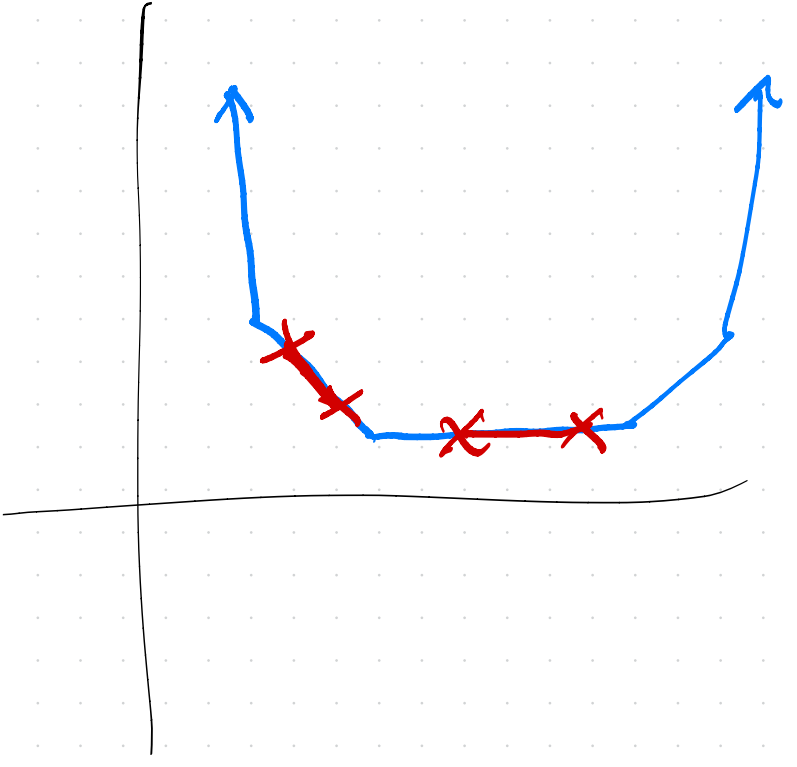
is not strictly convex!



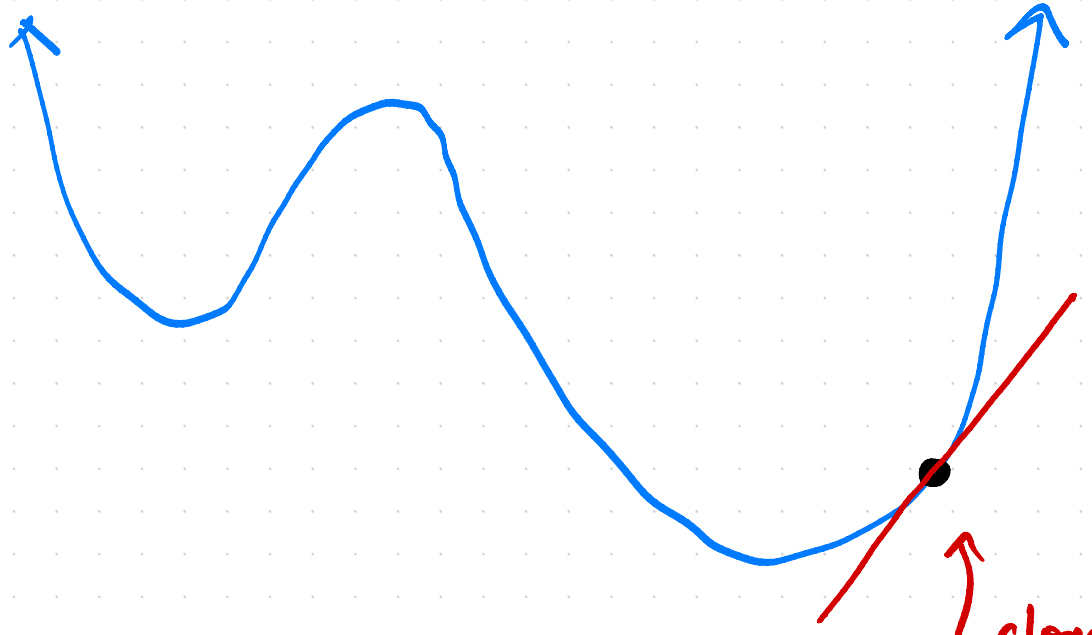
Strict convexity



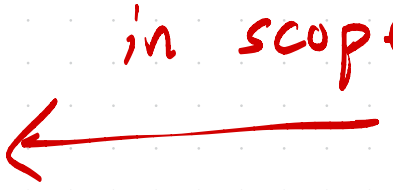
convex, but not strictly convex



Tangent line / plane / hyperplane



slope of tangent line
= derivative



not in scope for Midterm 2

(but definitely for the Final!)

Chapter 9: Eigenvalues and eigenvectors

(1) Better understand convexity and related ideas

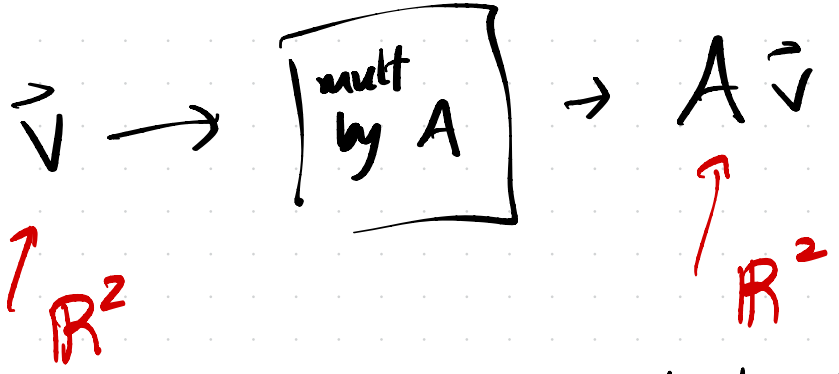
e.g. $X^T X + I$ is

always invertible, even though $X^T X$ isn't always: why?

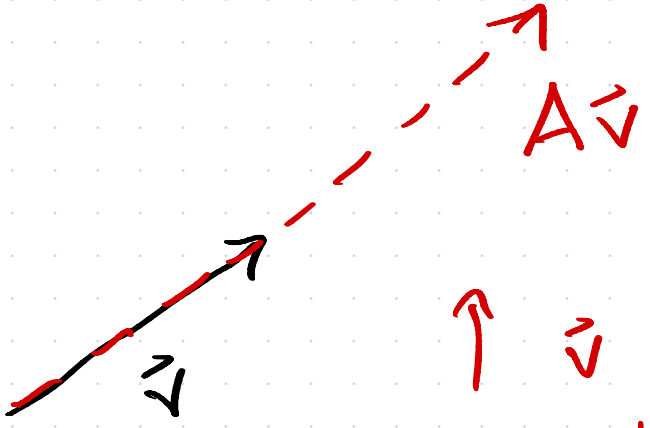
(2) Lots of new applications

Suppose A is an $n \times n$ matrix

e.g. A 2×2, $\vec{v} \in \mathbb{R}^2$



- rotation
- reflection
- projection
- shear
- stretch/compress



↑ $\vec{v}$ is an **eigenvector** of A,
because $A\vec{v}$ returns
a scalar multiple of $\vec{v}$

→ stretch factor: **eigenvalue**

Suppose A is an $n \times n$ matrix.

$\vec{v} \neq \vec{0}$ is an eigenvector of A, with eigenvalue λ, if

$$A\vec{v} = \lambda\vec{v}$$

interpretation: when multiplied by A, $\vec{v}$'s direction is unchanged.

$$A = \begin{bmatrix} 1 & 2 \\ 2 & 1 \end{bmatrix}$$

- Observe: $\vec{v} = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$ is an eigenvector!

$\lambda = 3$

$$A\begin{bmatrix} 1 \\ 1 \end{bmatrix} = \begin{bmatrix} 1 & 2 \\ 2 & 1 \end{bmatrix}\begin{bmatrix} 1 \\ 1 \end{bmatrix} = \begin{bmatrix} 3 \\ 3 \end{bmatrix} = 3\begin{bmatrix} 1 \\ 1 \end{bmatrix}$$

- So is $\begin{bmatrix} 15 \\ 15 \end{bmatrix}$!

also $\lambda = 3$!

$$A\begin{bmatrix} 15 \\ 15 \end{bmatrix} = \begin{bmatrix} 1 & 2 \\ 2 & 1 \end{bmatrix}\begin{bmatrix} 15 \\ 15 \end{bmatrix} = \begin{bmatrix} 45 \\ 45 \end{bmatrix} = 3\begin{bmatrix} 15 \\ 15 \end{bmatrix}$$

- So is any scalar multiple of $\begin{bmatrix} 1 \\ 1 \end{bmatrix}$

$$A = \begin{bmatrix} 1 & 2 \\ 2 & 1 \end{bmatrix}$$

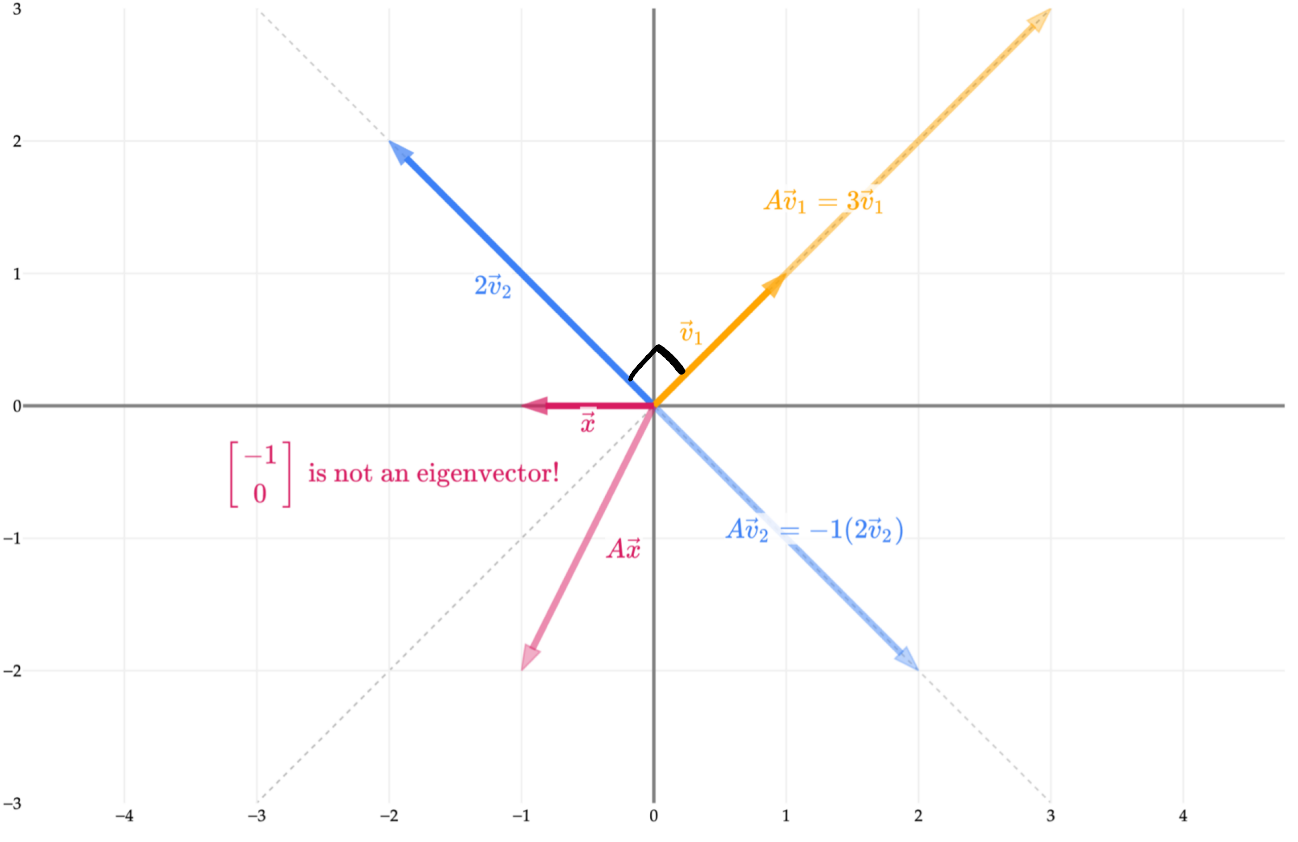
- Another eigenvector: $\begin{bmatrix} 1 \\ -1 \end{bmatrix}$

$$\begin{bmatrix} 1 & 2 \\ 2 & 1 \end{bmatrix} \begin{bmatrix} 1 \\ -1 \end{bmatrix} = \begin{bmatrix} -1 \\ 1 \end{bmatrix} = (-1) \begin{bmatrix} 1 \\ -1 \end{bmatrix}$$

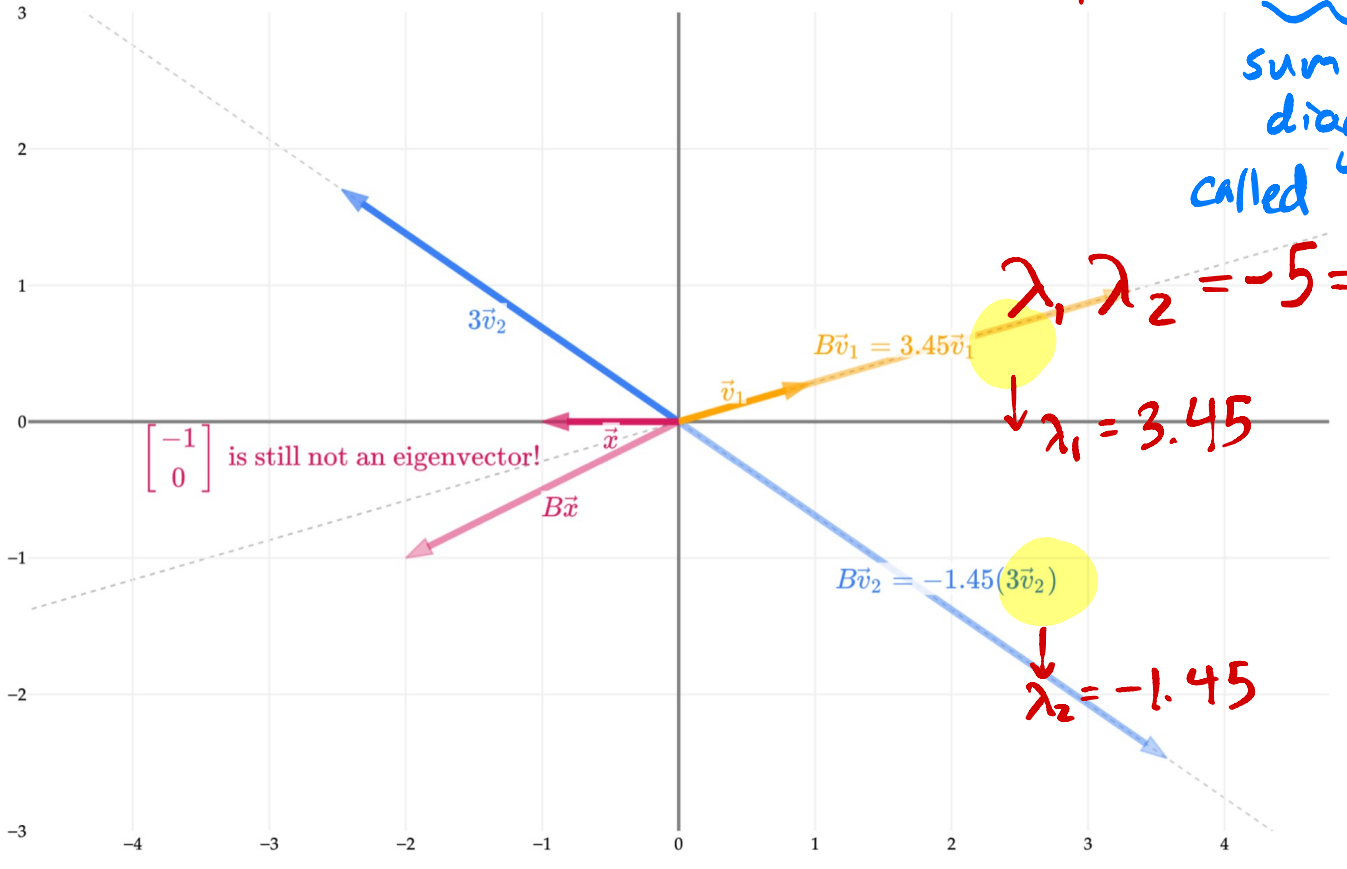
$A = \begin{bmatrix} 1 & 2 \\ 2 & 1 \end{bmatrix}$ $\quad \vec{v}_1 = \begin{bmatrix} 1 \\ 1 \end{bmatrix}, \lambda_1 = 3$

$\vec{v}_2 = \begin{bmatrix} 1 \\ -1 \end{bmatrix}, \lambda_2 = -1$

Visualizing the eigenvectors of $A = \begin{bmatrix} 1 & 2 \\ 2 & 1 \end{bmatrix}$



Visualizing the eigenvectors of $B = \begin{bmatrix} 2 & 5 \\ 1 & 0 \end{bmatrix}$



observe: $\lambda_1 + \lambda_2 = 2 + 0$

sum of diagonal, called "trace"

$\lambda_1 \lambda_2 = -5 =$ determinant

$\lambda_1 = 3.45$

$\lambda_2 = -1.45$

$$A = \begin{bmatrix} 1 & 4 \\ 3 & 12 \end{bmatrix}$$

What are its eigenvalues and eigenvectors?

$4\vec{a}^{(1)} = \vec{a}^{(2)}$

$-4\vec{a}^{(1)} + \vec{a}^{(2)} = \vec{0}$

Observe: $\text{rank}(A) = 1$, so $\det(A) = 0$

$\vec{v}_2 = \begin{bmatrix} 1 \\ 3 \end{bmatrix}$

$\lambda_1 = 0$

$\vec{v}_1 = \begin{bmatrix} -4 \\ 1 \end{bmatrix}$

$\vec{v}_1 \in \text{nullsp}(A)$!

$\det(A) = 0 \iff 0$ is an eigenvalue

$\lambda_2 = 13$

$$\begin{bmatrix} 1 & 4 \\ 3 & 12 \end{bmatrix} \begin{bmatrix} a \\ b \end{bmatrix} = \begin{bmatrix} 13a \\ 13b \end{bmatrix}$$

$a + 4b = 13a \Rightarrow 4b = 12a \Rightarrow b = 3a$

$3a + 12b = 13b \rightarrow 3a = b$

infinitely many a, b! pick any pair, e.g. $a = 1, b = 3$

Finding eigenvalues in general

let's find a condition for λ !

$$A\vec{v} = \lambda\vec{v}$$

$$(A - \lambda I)\vec{v} = \vec{0}$$

says $\vec{v} \in \text{nullsp}(A - \lambda I)$!

but that means $A - \lambda I$ is not invertible!

$$\Rightarrow \quad \det(A - \lambda I) = 0 \ !$$

"characteristic polynomial" $p(\lambda)$

$$A = \begin{bmatrix} 1 & 2 \\ 2 & 1 \end{bmatrix}$$

$$A - \lambda I = \begin{bmatrix} 1-\lambda & 2 \\ 2 & 1-\lambda \end{bmatrix}$$

$$\det(A - \lambda I) = \underbrace{(1-\lambda)^2 - 4}_{p(\lambda)}$$